

TE 452 / TE 462 - Telecommunications Policy & Regulation

200 Multiple-Choice Questions with Answers

Part 1

1. A mobile operator lowers the monthly price of its data bundle and observes that the total quantity of bundles subscribers buy rises. Which principle does this most directly illustrate?

- A. The law of supply, where quantity supplied rises as price falls in the short run
- B. The law of demand, where quantity demanded rises as price falls, other things equal ✓**
- C. A rightward shift of the demand curve caused by the price reduction
- D. An increase in demand elasticity that automatically raises total revenue

Answer: B. The law of demand, where quantity demanded rises as price falls, other things equal

2. An analyst notes that the quantity of broadband demanded changed solely because the price of broadband itself changed, with incomes and tastes unchanged. This is best described as:

- A. A rightward shift of the entire demand curve
- B. A movement along the existing demand curve ✓**
- C. A leftward shift of the entire demand curve
- D. A change in the position of the supply curve

Answer: B. A movement along the existing demand curve

3. When the price of fresh yoghurt rises sharply, many buyers switch to alternatives such as Fanyogo or Fanice instead. In demand analysis these alternatives are best classified as:

- A. Complementary goods
- B. Substitute goods ✓**
- C. Inferior goods
- D. Giffen goods

Answer: B. Substitute goods

4. Among the many determinants of the quantity demanded of a good, economic theory identifies one variable as central to the basic demand relationship. That variable is the:

- A. Number of sellers operating in the market
- B. Cost of the inputs used in production
- C. Level of technology available to producers
- D. Price of the good itself ✓**

Answer: D. Price of the good itself

5. At higher prices a telecom equipment vendor finds it profitable to offer more units, while at lower prices it offers fewer. This relationship between price and quantity offered is captured by the:

- A. Law of demand
- B. Law of supply ✓**
- C. Law of diminishing marginal utility
- D. Principle of tax incidence

Answer: B. Law of supply

6. The market supply curve is drawn holding input costs constant. If the price of a key input such as sugar falls for ice-cream makers, the most likely result is that the supply curve:

- A. Shifts to the left, as producers supply less at each price
- B. Becomes perfectly vertical at the equilibrium quantity
- C. Stays fixed while only quantity demanded increases

D. Shifts to the right, as producers supply more at each price ✓

Answer: D. Shifts to the right, as producers supply more at each price

7. Demand for a particular good is described as elastic. This means that:

A. Quantity demanded responds substantially to a change in the good's price ✓

B. Quantity demanded barely responds to a change in the good's price

C. Quantity supplied responds substantially to a change in the good's price

D. Price remains fixed regardless of changes in quantity demanded

Answer: A. Quantity demanded responds substantially to a change in the good's price

8. A regulator studies a service that consumers treat as a necessity with few substitutes, so a price increase changes the quantity bought only slightly. Demand for this service is best described as:

A. Elastic

B. Inelastic ✓

C. Perfectly elastic

D. Unit elastic

Answer: B. Inelastic

9. On a standard supply-and-demand diagram, the single point at which the supply curve and the demand curve intersect determines the:

A. Price ceiling and the resulting shortage

B. Price floor and the resulting surplus

C. Tax incidence borne by buyers and sellers

D. Equilibrium price and equilibrium quantity ✓

Answer: D. Equilibrium price and equilibrium quantity

10. To keep a service affordable, a government sets a maximum legal price below the market equilibrium price. Standard analysis predicts this price ceiling will most likely create:

A. A surplus, because quantity supplied exceeds quantity demanded

B. A shortage, because quantity demanded exceeds quantity supplied ✓

C. A new stable equilibrium exactly at the controlled price

D. An increase in supply that quickly eliminates excess demand

Answer: B. A shortage, because quantity demanded exceeds quantity supplied

11. A government sets a minimum legal price above the market equilibrium to protect suppliers. The most likely consequence of this price floor is:

A. A shortage, because quantity demanded exceeds quantity supplied

B. An immediate return of the market to its equilibrium

C. A reduction in quantity supplied below the quantity demanded

D. A surplus, because quantity supplied exceeds quantity demanded ✓

Answer: D. A surplus, because quantity supplied exceeds quantity demanded

12. In public finance, the term tax incidence refers specifically to:

A. The statutory tax rate that a government formally sets out in legislation

B. The total amount of revenue that a government collects from levying the tax

C. How the burden of a tax is distributed among the participants in a market ✓

D. The administrative cost a government incurs in collecting and enforcing the tax

Answer: C. How the burden of a tax is distributed among the participants in a market

13. When a per-unit tax is imposed on the sellers of a good, the standard model represents its initial effect as:

A. A rightward shift of the supply curve

B. A leftward shift of the demand curve

C. A leftward (upward) shift of the supply curve ✓

D. A rotation that makes the demand curve more elastic

Answer: C. A leftward (upward) shift of the supply curve

14. Two markets face an identical per-unit tax. In Market A demand is highly inelastic; in Market B demand is highly elastic. Compared with sellers, buyers will tend to bear the larger share of the tax in:

- A. Market B, where demand is elastic
- B. Market A, where demand is inelastic ✓**
- C. Both markets equally, regardless of elasticity
- D. Neither market, because sellers always remit the tax

Answer: B. Market A, where demand is inelastic

15. An economic system in which production and allocation follow long-standing customary patterns and local leaders have the ultimate say is best described as a:

- A. Command economy, directed by a central government authority
- B. Market economy, coordinated primarily through prices and competition
- C. Traditional economy, based on inherited customs and social roles ✓**
- D. Mixed economy, blending market forces with state intervention

Answer: C. Traditional economy, based on inherited customs and social roles

16. Twentieth-century experience suggests that a command economy tends to fail largely because:

- A. Prices adjust too quickly for producers to keep pace with rapidly shifting demand
- B. Private firms refuse to compete once the state withdraws support and guaranteed contracts
- C. A central authority cannot efficiently gather and act on the information needed to allocate resources ✓**
- D. Consumers are given so many product choices that planners cannot coordinate production

Answer: C. A central authority cannot efficiently gather and act on the information needed to allocate resources

17. A mixed economy is best characterised as one in which:

- A. A central planner sets all prices, wages and output levels across the whole economy
- B. Custom and tradition determine nearly all economic decisions, with no market pricing
- C. Market forces operate freely with no government involvement of any kind whatsoever
- D. Markets allocate most resources while governments intervene to correct market failures ✓**

Answer: D. Markets allocate most resources while governments intervene to correct market failures

18. Advocates argue that a market economy can be self-optimising because, when individuals are free to pursue their own interests, the system tends to:

- A. Eliminate the need for any property rights, contracts or enforcement mechanisms
- B. Guarantee an equal distribution of income across the whole of society
- C. Prevent monopolies from ever arising in any industry whatsoever
- D. Settle at a stable equilibrium of supply, consumption and prices ✓**

Answer: D. Settle at a stable equilibrium of supply, consumption and prices

19. In a perfectly competitive market each individual buyer and seller is assumed to be a price taker. This assumption follows from the fact that:

- A. A single dominant firm sets the price that every other participant follows
- B. The government fixes the price at which every trade in the market must occur
- C. There are so many participants that no single one can influence the market price ✓**
- D. Buyers and sellers openly collude with one another to maintain a single common price

Answer: C. There are so many participants that no single one can influence the market price

20. Compared with a competitive industry, a profit-maximising monopoly is generally expected to:

- A. Produce more output and charge a lower price
- B. Produce less output and charge a higher price ✓**
- C. Produce the same output but at a lower cost

D. Charge a lower price while expanding total output

Answer: B. Produce less output and charge a higher price

21. An industry is termed a natural monopoly when:

A. Two or more competing firms can always supply the market more cheaply than one

B. The government simply bans all but one licensed firm from entering the market

C. Firms in the industry naturally collude to behave as though they were one

D. A single firm can supply the whole market at lower cost than two or more firms could ✓

Answer: D. A single firm can supply the whole market at lower cost than two or more firms could

22. Where a genuine natural monopoly exists, a government wishing to avoid wasteful duplication of infrastructure may reasonably choose to:

A. Tolerate the single firm but subject it to regulation ✓

B. Force several firms to compete regardless of cost

C. Withdraw entirely and leave the market unregulated

D. Nationalise every competing firm in the wider economy

Answer: A. Tolerate the single firm but subject it to regulation

23. Within the logic of a market economy, the primary justification for governments intervening through regulators and other institutions is that:

A. Markets can fail, and intervention may correct such failures ✓

B. Markets always allocate resources perfectly without help

C. Government planning is inherently superior to all markets

D. Competition should be eliminated to simplify oversight

Answer: A. Markets can fail, and intervention may correct such failures

24. An economist distinguishes a change in demand from a change in quantity demanded. A change in demand (a shift of the curve) is caused by:

A. A change in the good's own price alone, holding all else constant

B. A movement to a different point along the same, unchanged demand curve

C. A change in a determinant other than the good's own price, such as income or tastes ✓

D. A simultaneous change in supply that happens to exactly offset any change in demand

Answer: C. A change in a determinant other than the good's own price, such as income or tastes

25. One reason a monopoly is considered socially harmful is that its output is typically produced:

A. More efficiently and at a lower cost than under competition

B. Less efficiently and at a higher cost than under competition ✓

C. At exactly the competitive level of cost and efficiency

D. Only when it is directly subsidised by the government

Answer: B. Less efficiently and at a higher cost than under competition

26. Before the 1980s, telecommunications services in most countries were typically provided by:

A. Numerous competing private operators

B. Independent regional cooperatives

C. State-owned monopolies ✓

D. Foreign-owned multinational consortia

Answer: C. State-owned monopolies

27. A factor underpinning the pre-1980s telecom monopoly paradigm was economic: high network build costs and low demand meant a second operator could not easily recover its costs. This gave the industry the character of a:

A. Natural monopoly ✓

B. Perfectly contestable market

C. Fully deregulated free market

D. Command-economy consumer good

Answer: A. Natural monopoly

28. A political factor behind early state telecom monopolies was that governments:

- A. Were legally barred by international postal treaty from ever licensing a private telecom firm
- B. Lacked any government ministry with the technical capacity to run even a basic postal service
- C. Preferred to auction off spectrum rights directly to the highest-bidding private investor

D. Wished to retain state control over communications, often treating them as an extension of postal networks ✓

Answer: D. Wished to retain state control over communications, often treating them as an extension of postal networks

29. In telecommunications, liberalization is best defined as:

- A. The transfer of an operator's ownership from the state to private hands
- B. The intervention of government to achieve specific market outcomes
- C. The process of transition from a monopoly market to a competitive one ✓**
- D. The setting of technical standards common to all operators

Answer: C. The process of transition from a monopoly market to a competitive one

30. A central element of liberalization is the removal of exclusivities from the former monopoly fixed-network operator, commonly called the incumbent. The direct effect of removing these exclusivities is to:

- A. Allow new entrants to compete in services that were previously monopolised ✓**
- B. Transfer ownership of the incumbent's shares to a group of private investors
- C. Guarantee the incumbent a permanently protected and exclusive market position
- D. Impose strict price caps on every single service the incumbent operator offers

Answer: A. Allow new entrants to compete in services that were previously monopolised

31. Privatization in the telecom sector refers to:

- A. The transition of an entire market from monopoly toward open competition
- B. The imposition of universal-service obligations on all licensed operators
- C. The transfer of ownership of an operator from the state to the private sector ✓**
- D. The creation of a wholly independent national regulatory authority

Answer: C. The transfer of ownership of an operator from the state to the private sector

32. Governments have privatized incumbent operators in different ways. Selling a stake to a single strategic investor, rather than via a broad public share sale, has been the more common approach in:

- A. Highly industrialised countries
- B. Countries that have no incumbent operator
- C. Countries that retained a full state monopoly

D. Developing countries ✓

Answer: D. Developing countries

33. A country opens its mobile market to several competing firms but keeps the incumbent fixed-line operator fully state-owned. This situation is best described as:

- A. Privatization of the incumbent operator without any market liberalization
- B. Liberalization of the market without privatization of the incumbent operator ✓**
- C. Both full market liberalization and full privatization of the incumbent
- D. Neither market liberalization nor privatization of the incumbent at all

Answer: B. Liberalization of the market without privatization of the incumbent operator

34. In the context of the telecom sector, regulation is best defined as:

- A. Government intervention in markets to achieve certain desired outcomes ✓**
- B. The complete and permanent removal of government from all market activity
- C. The wholesale transfer of state-owned assets into full private ownership

D. The voluntary self-policing of firms with absolutely no external oversight

Answer: A. Government intervention in markets to achieve certain desired outcomes

35. Regulation is sometimes said to take three basic forms. The form that establishes a common, legally enforceable platform for all businesses, such as rules governing electrical supply or company accounting, is:

A. Competition regulation

B. Public policy regulation

C. Technical standardization ✓

D. Voluntary self-regulation

Answer: C. Technical standardization

36. Legislation that controls the supply of goods such as alcohol, gambling or pornography because doing so is believed to serve the public interest is an example of:

A. Technical standardization

B. Competition regulation

C. Public policy regulation ✓

D. Co-regulation

Answer: C. Public policy regulation

37. The form of regulation that seeks to create and maintain fair competitive relationships between businesses, and between businesses and consumers, is:

A. Technical standardization

B. Competition regulation ✓

C. Public policy regulation

D. Zero regulation

Answer: B. Competition regulation

38. Although telecom regulation can take all three basic forms, the principal focus of contemporary regulation of the telecommunications services industry is:

A. The standardisation of accounting rules across all industries

B. The control of socially sensitive goods such as gambling

C. The creation, nourishment and maintenance of competitive markets ✓

D. The elimination of private ownership from the sector

Answer: C. The creation, nourishment and maintenance of competitive markets

39. Among the widely accepted regulatory objectives in the telecom sector is the aim to:

A. Guarantee a fixed profit margin to every licensed operator

B. Restrict connectivity to commercially attractive urban areas

C. Promote universal access to basic telecommunications services ✓

D. Prevent any new operators from entering the market

Answer: C. Promote universal access to basic telecommunications services

40. A recognised regulatory objective is that, where competitive markets do not exist or have failed, the regulator should:

A. Guarantee the dominant firm full protection from all potential competitors

B. Prevent abuses of market power such as excessive pricing by dominant firms ✓

C. Withdraw all regulatory oversight and let the market correct itself unaided

D. Fix every operator's retail prices at exactly identical levels nationwide

Answer: B. Prevent abuses of market power such as excessive pricing by dominant firms

41. Which statement best captures the distinction between policy and law as drawn in the course material?

A. Law can compel or prohibit behaviour, whereas policy merely guides actions toward a desired outcome ✓

B. Policy can compel or prohibit behaviour, whereas law only ever offers non-binding guidance

- C. Policy and law are identical concepts, so the two terms may always be used interchangeably
- D. Regulation sets the broad long-term goals while policy enforces them through the courts

Answer: A. Law can compel or prohibit behaviour, whereas policy merely guides actions toward a desired outcome

42. In the policy-to-practice chain, regulation is primarily concerned with:

- A. Translating the objectives of policy decisions into practical rules and decisions ✓**
- B. Setting the broad, long-term national vision that exists prior to any policy
- C. Replacing the need for any underlying legislation or statutory authority
- D. Auditing the personal finances of individual government ministers each year

Answer: A. Translating the objectives of policy decisions into practical rules and decisions

43. In many countries before reform, a single ministry or government unit often acted simultaneously as:

- A. An independent regulator that is fully separated from the operator
- B. A purely commercial firm with no policy responsibilities whatsoever
- C. Policy maker, policy implementer and operator of the telephone service ✓**
- D. A neutral standards body with no operational role in the sector

Answer: C. Policy maker, policy implementer and operator of the telephone service

44. The International Telecommunication Union (ITU) is best described as:

- A. A private consortium formed and funded by the world's largest mobile network operators worldwide
- B. A regional trade bloc that negotiates import tariffs on telecommunications equipment for its members
- C. A specialised agency of the United Nations responsible for information and communication technology issues ✓**
- D. A commercial standards-setting firm that is owned and controlled by major equipment vendors

Answer: C. A specialised agency of the United Nations responsible for information and communication technology issues

45. Within the ITU's three-sector structure, the sector that manages the international radio-frequency spectrum and satellite-orbit resources is:

- A. ITU-R (Radiocommunication) ✓**
- B. ITU-T (Standardization)
- C. ITU-D (Development)
- D. ITU-P (Policy)

Answer: A. ITU-R (Radiocommunication)

46. The ITU sector responsible for standardizing global telecommunications, except for radio matters, is:

- A. ITU-R (Radiocommunication)
- B. ITU-D (Development)
- C. ITU-T (Standardization) ✓**
- D. ITU-S (Spectrum)

Answer: C. ITU-T (Standardization)

47. Helping to spread equitable, sustainable and affordable access to ICTs is the principal role of which ITU sector?

- A. ITU-R (Radiocommunication)
- B. ITU-D (Development) ✓**
- C. ITU-T (Standardization)
- D. ITU-A (Access)

Answer: B. ITU-D (Development)

48. The World Trade Organization's Agreement on Basic Telecommunications (the BTA) primarily:

- A. Allocates radio spectrum directly to individual telecom operators in each member state

B. Establishes rules for competition and promotes open investment in telecoms across countries ✓

C. Sets the maximum retail tariffs that operators may charge their consumers directly

D. Issues operating licences directly to individual carriers within each member state

Answer: B. Establishes rules for competition and promotes open investment in telecoms across countries

49. One of the six basic principles of the WTO Reference Paper on Regulation is that:

A. The dominant operator should also serve as the regulator

B. Licensing criteria should be kept confidential from competitors

C. Scarce resources should be reserved for the incumbent operator

D. There should be a regulatory body independent of operating companies ✓

Answer: D. There should be a regulatory body independent of operating companies

50. The WTO Reference Paper requires telecom regulators to be independent of operating companies. On independence from government, the material indicates that regulators are:

A. Not necessarily independent of governments, though greater independence is often regarded as better ✓

B. Required by law to be wholly controlled and directed by the relevant government ministry

C. Required by law to be completely and formally independent of government in every respect

D. Required by law to be jointly owned and controlled by the operators that they regulate

Answer: A. Not necessarily independent of governments, though greater independence is often regarded as better

51. According to the framework for regulation, the laws passed by governments to enable regulation typically:

A. Guarantee permanent monopoly status to the incumbent state operator

B. Appoint regulatory bodies and confer on them powers to take action ✓

C. Remove all accountability and reporting requirements from regulators

D. Fix the retail prices of every single regulated service nationwide

Answer: B. Appoint regulatory bodies and confer on them powers to take action

52. Which option best lists the categories of law that bear specifically on the telecommunications services industry?

A. Sector-specific telecom law, competition and fair-trading law, and other relevant law such as privacy ✓

B. Criminal law, international maritime law and constitutional law only, excluding all else

C. Tax law, general employment law and intellectual-property law only, and nothing further

D. Banking law, insurance law and environmental law only, excluding any other category

Answer: A. Sector-specific telecom law, competition and fair-trading law, and other relevant law such as privacy

53. Provisions enacted by bodies such as ECOWAS or the WTO take force in a member or signatory country when they are:

A. Transposed into national law in accordance with treaty obligations ✓

B. Simply published on the regional body's official website

C. Approved by a majority vote of the country's licensed operators

D. Adopted automatically the moment they are agreed upon regionally

Answer: A. Transposed into national law in accordance with treaty obligations

54. The framework notes that the function of regulation may be split between two or more bodies. A risk a country must manage when doing so is that:

A. Poorly designed division of functions can lead to disagreements and slow decision-making ✓

B. A single national regulator will inevitably end up capturing the entire market

C. Operators will find themselves completely unable to obtain any licence at all

D. Spectrum will automatically be allocated to all operators entirely free of charge

Answer: A. Poorly designed division of functions can lead to disagreements and slow decision-making

55. Listed in increasing order of intrusiveness, the styles of regulation are:

- A. Formal regulation, co-regulation, self-regulation, zero regulation
- B. Zero regulation, self-regulation, co-regulation, formal regulation ✓**
- C. Self-regulation, zero regulation, formal regulation, co-regulation
- D. Co-regulation, formal regulation, zero regulation, self-regulation

Answer: B. Zero regulation, self-regulation, co-regulation, formal regulation

56. An industry described as operating under zero regulation is one in which:

- A. A regulator intervenes minutely in the firm's daily operations
- B. Firms jointly agree binding rules under an industry watchdog
- C. Prices are set directly and solely by a government ministry
- D. Market forces are allowed to work with no external intervention ✓**

Answer: D. Market forces are allowed to work with no external intervention

57. Self-regulation, in which the players in an industry regulate themselves, tends to be MOST effective when:

- A. There is a high convergence of interest among the stakeholders ✓**
- B. There are strong commercial conflicts between the players
- C. A single firm dominates and distrusts all the others
- D. The regulator intervenes in every operational decision

Answer: A. There is a high convergence of interest among the stakeholders

58. A frequently cited benefit of self-regulation is that:

- A. Companies usually understand their own businesses better than an external regulator does ✓**
- B. It removes all administrative costs of regulation from consumers and industry
- C. It guarantees that no company will ever act opportunistically toward rivals
- D. It eliminates the need for any underlying legislation or licensing regime

Answer: A. Companies usually understand their own businesses better than an external regulator does

59. Co-regulation is best described as an arrangement in which:

- A. Firms operate entirely independently with no external oversight of any kind whatsoever
- B. The regulator alone dictates every technical and pricing decision unilaterally
- C. The regulator joins industry players in a joint forum rather than regulating purely from outside ✓**
- D. A foreign international body simply assumes all domestic regulatory authority

Answer: C. The regulator joins industry players in a joint forum rather than regulating purely from outside

60. While potentially very successful, co-regulation is particularly vulnerable to:

- A. An immediate and total loss of all formal regulatory powers
- B. The automatic capture of the regulator by consumer groups
- C. Role ambiguity and differing perceptions of what each player will do ✓**
- D. A blanket legal prohibition on any form of industry collaboration

Answer: C. Role ambiguity and differing perceptions of what each player will do

61. Regulation is described as intrusive when the regulator:

- A. Lays down only general principles and leaves the methods to firms themselves
- B. Withdraws entirely from the market and lets market forces operate unaided
- C. Intervenes often and minutely in the day-to-day running of a company's business ✓**
- D. Limits itself only to publishing occasional, non-binding guidelines

Answer: C. Intervenes often and minutely in the day-to-day running of a company's business

62. One reason intrusive regulation is considered undesirable is that:

- A. It always sharply reduces the total administrative cost of regulating the industry
- B. It guarantees that competitive market forces will always prevail entirely unaided
- C. The regulator effectively becomes a proxy manager of the industry, weakening operators' freedom ✓**
- D. It completely removes the regulator's responsibility for any market outcomes at all

Answer: C. The regulator effectively becomes a proxy manager of the industry, weakening operators' freedom

63. A good regulator is advised to make regulation pervasive rather than intrusive. This means regulation should:

- A. Influence operators broadly while letting them manage their own affairs within the spirit of the rules ✓**
- B. Dictate the exact, precise technical solution required of every single licensed operator
- C. Have as little practical impact as possible on any of the companies operating in the market
- D. Apply narrowly to only a single dominant operator in the market, ignoring all the rest

Answer: A. Influence operators broadly while letting them manage their own affairs within the spirit of the rules

64. The main instruments through which regulation is accomplished include authorizations, rules, determinations, consents and:

- A. Guidelines ✓**
- B. Share buy-backs
- C. Spectrum auctions
- D. Corporate tax assessments

Answer: A. Guidelines

65. A system of authorization may issue specific authorizations and general authorizations. These correspond respectively to:

- A. Spectrum licences and national numbering plans
- B. Individual operator licences and class licences ✓**
- C. Regulatory price caps and interconnection orders
- D. Formal consents and regulatory determinations

Answer: B. Individual operator licences and class licences

66. Determinations are instruments of ongoing regulation that, in effect:

- A. Permanently revoke an operator's licence without any right of appeal
- B. Replace the underlying legislation entirely and without warning
- C. Elaborate the rules, for example in the periodic review of price controls ✓**
- D. Carry no legal force at all and are merely advisory in nature

Answer: C. Elaborate the rules, for example in the periodic review of price controls

67. A consent is best understood as:

- A. An order that formally imposes an additional new obligation on the operator concerned
- B. An informal guideline document that carries no legal status whatsoever in practice
- C. A formal financial penalty levied against an operator for breach of a licence condition
- D. A negative determination that formally relieves an operator from complying with a requirement ✓**

Answer: D. A negative determination that formally relieves an operator from complying with a requirement

68. Guidelines differ from rules and determinations in that guidelines:

- A. Have far greater binding legal force than the licence conditions do themselves
- B. Do not in themselves possess legal status and are usually written in user-friendly language ✓**
- C. Can be issued only by the courts, and only following a successful appeal process
- D. Automatically and permanently revoke a licence the very moment they are breached

Answer: B. Do not in themselves possess legal status and are usually written in user-friendly language

69. A key reason regulators consult widely before finalising instruments is that consultation:

- A. Transfers all formal decision-making power directly to the consulted parties
- B. Entirely removes the regulator's legal obligation to justify its decisions
- C. Gives industry ownership and pre-knowledge of decisions and enriches decision-making ✓**
- D. Guarantees that no consulted party will ever appeal the final regulatory outcome

Answer: C. Gives industry ownership and pre-knowledge of decisions and enriches decision-making

70. An appeal against a regulator's action may be brought on the grounds of law (illegality), procedure (misadministration) or:

- A. Profitability (commercial loss)
- B. Logic (irrationality) ✓**
- C. Popularity (public disapproval)
- D. Convenience (administrative delay)

Answer: B. Logic (irrationality)

71. Which of the following is a recognised enforcement power available to a regulator dealing with rule breaches?

- A. Seizing ownership of a competitor's shares
- B. Setting the personal salaries of company directors
- C. Revoking a licence, or imposing fines and compliance orders ✓**
- D. Banning the company's customers from switching providers

Answer: C. Revoking a licence, or imposing fines and compliance orders

72. Regarding the funding of national regulatory authorities, the typical arrangement is that:

- A. Operators are always legally barred from contributing any funds to the regulator at all
- B. Regulators are funded entirely by fines collected from ordinary telecom consumers
- C. Funding comes entirely from international donor agencies and foreign aid grants
- D. Governments assume primary responsibility but usually pass the cost onto industry through fees ✓**

Answer: D. Governments assume primary responsibility but usually pass the cost onto industry through fees

73. The concept of regulatory hazard refers to the fact that:

- A. Regulators personally face genuine physical danger when inspecting live telecom networks
- B. Regulation always removes every last trace of risk from the telecom market entirely
- C. Operators face absolutely no consequences whatsoever for repeatedly breaching the rules
- D. Regulatory power, if exercised unwisely, can distort markets and damage the industry it serves ✓**

Answer: D. Regulatory power, if exercised unwisely, can distort markets and damage the industry it serves

74. Regulatory capture occurs when:

- A. The regulator becomes completely paralysed and simply stops issuing any decisions at all
- B. Individual operators are captured and then run directly by the regulator itself as owner
- C. Ordinary consumers themselves take over the entire day-to-day running of the regulator
- D. The regulator identifies too closely with, and so serves the interests of, the industry it regulates ✓**

Answer: D. The regulator identifies too closely with, and so serves the interests of, the industry it regulates

75. The information asymmetry, or knowledge gap, that regulators face arises because:

- A. Regulators are always legally forbidden from collecting any operational data at all
- B. Consumers deliberately conceal their true preferences from the regulator entirely
- C. Operators are always legally required to share none of their internal financial records

D. The companies being regulated usually know far more about the business than the regulator does ✓

Answer: D. The companies being regulated usually know far more about the business than the regulator does

76. Regulators issue licences to companies primarily to authorise them to:

A. Operate telecommunications networks and provide telecommunications services ✓

B. Manufacture and export telecommunications equipment internationally

C. Set the national monetary and fiscal policy for the entire economy

D. Audit the financial accounts of every single rival operator each year

Answer: A. Operate telecommunications networks and provide telecommunications services

77. Beyond granting permission to operate, a telecommunications licence characteristically specifies:

A. The exact retail prices that every other licensed operator in the country must charge consumers

B. The complete internal staffing structure and organisational chart of the regulator itself

C. What facilities a company may deploy, what services it may offer, and how it must interact with other businesses ✓

D. The personal income tax obligations of each of the company's individual board directors

Answer: C. What facilities a company may deploy, what services it may offer, and how it must interact with other businesses

78. In addition to facilities and services, major licences may govern access to scarce resources such as spectrum and numbers, and commonly require companies to pay:

A. Upfront fees when the licence is awarded and annual payments thereafter ✓

B. A fixed share of annual profits paid to every competing operator

C. Nothing at all, since licences are always granted completely free of charge

D. A single one-time deposit that is always fully refunded upon expiry

Answer: A. Upfront fees when the licence is awarded and annual payments thereafter

79. Historically, telecom licences were allocated separately for different sub-markets, for example fixed networks, mobile networks or paging. The main characteristic of this approach was that licences:

A. Were completely technology- and service-neutral by design

B. Imposed identical licence terms across every single market segment

C. Differentiated between specific technologies and services ✓

D. Were never tied to any particular type of technology or service

Answer: C. Differentiated between specific technologies and services

80. Recent technological convergence, which lets different platforms deliver a wider range of services, has prompted some regulators to move toward:

A. Stricter technology-specific licensing for each platform

B. Technology- and service-neutral licensing ✓

C. The complete abolition of all forms of licensing

D. Permanent exclusive licences for incumbent operators

Answer: B. Technology- and service-neutral licensing

81. A technology-neutral licence to provide a telephone service is one that:

A. Requires the company to use one single, specified fixed-line technology only

B. Strictly forbids the company from offering any form of voice service at all

C. Mandates the exclusive use of a single, government-approved equipment vendor only

D. Lets the company choose whether to use a fixed network or a particular wireless technology ✓

Answer: D. Lets the company choose whether to use a fixed network or a particular wireless technology

82. The two main types of licence awarded to telecom companies are:

- A. Spectrum-specific licences and separate telephone numbering licences
- B. International import licences and international export licences
- C. Short-term provisional licences and longer-term probationary licences
- D. Individual operator licences and class licences (general authorizations) ✓**

Answer: D. Individual operator licences and class licences (general authorizations)

83. Individual operator licences are used primarily where:

- A. The market segment has essentially unlimited room for any number of new entrants to join
- B. Absolutely no scarce resources of any kind whatsoever are ever involved in that segment
- C. Perfectly identical terms are wanted for every single business operating in that market segment
- D. The number of licences is limited, dominance is likely, or specific resources such as spectrum must be allocated ✓**

Answer: D. The number of licences is limited, dominance is likely, or specific resources such as spectrum must be allocated

84. Class licences (general authorizations) are characterised by the fact that they:

- A. Are always tailored individually to just one single, specifically named operator
- B. Set out identical terms of operation for all businesses in a market segment, such as ISPs ✓**
- C. Are reserved exclusively for use by the former state incumbent operator
- D. Apply only in the specific cases where radio spectrum must be allocated

Answer: B. Set out identical terms of operation for all businesses in a market segment, such as ISPs

85. A current trend favours replacing individual operator licences with general authorizations wherever possible because doing so:

- A. Simplifies regulation, helps ensure a level playing field, and is more flexible to market change ✓**
- B. Maximises the one-time upfront revenue collected from each and every individual operator
- C. Permanently guarantees the incumbent a fully protected monopoly position
- D. Completely eliminates the need for any independent regulator in the sector whatsoever

Answer: A. Simplifies regulation, helps ensure a level playing field, and is more flexible to market change

86. One licensing objective treats basic telecommunications as an essential public service. This justifies:

- A. Removing all regulatory oversight the moment the market is privatised
- B. Retaining some controls to ensure services are provided in the public interest ✓**
- C. Reserving service provision for a single state monopoly forever
- D. Allowing operators to serve only the most profitable customers

Answer: B. Retaining some controls to ensure services are provided in the public interest

87. Network roll-out and service-coverage obligations are often written into licences to advance which licensing objective?

- A. Generation of one-time government spectrum auction revenue
- B. Protection of the incumbent operator from any competition
- C. Standardisation of company accounting and reporting practices
- D. Expansion of networks and services, and universal service ✓**

Answer: D. Expansion of networks and services, and universal service

88. When a state-owned incumbent is privatized, a licence is necessary because it:

- A. Transfers full legal ownership and operational control of the regulator to the investor
- B. Specifies what the investor is buying and what the government expects from the operator ✓**
- C. Guarantees the investor complete and permanent freedom from all regulation
- D. Entirely removes the operator's obligation to serve any customers at all

Answer: B. Specifies what the investor is buying and what the government expects from the operator

89. Licensing can be used to regulate market structure, most directly by:

- A. Fixing the wholesale price of every single network input
 - B. Determining the number of operators licensed to provide services ✓**
 - C. Setting the wages paid to all of an operator's employees
 - D. Selecting which equipment vendors operators must always use
- Answer: B. Determining the number of operators licensed to provide services**

90. Licence conditions intended to create a level playing field and limit incumbents' abuse of dominance are generally referred to as:

- A. Universal-service obligations for underserved regions
 - B. Spectrum-refarming and reallocation provisions
 - C. Tariff-rebalancing and price-adjustment schedules
 - D. Anti-competitive safeguards or fair-trading conditions ✓**
- Answer: D. Anti-competitive safeguards or fair-trading conditions**

91. Finite resources such as radio spectrum, numbers and rights of way should, as a licensing objective, be allocated:

- A. Entirely to whichever operator has the longest operating history
 - B. Purely on a first-come basis with no regard to efficiency at all
 - C. Only to operators that are owned by the national government
 - D. Between operators fairly, efficiently and in the public interest ✓**
- Answer: D. Between operators fairly, efficiently and in the public interest**

92. Licensing can generate government revenue in two characteristic ways:

- A. One-time revenues from auctions and continuing revenue from annual licence fees ✓**
 - B. Confiscation of operator profits and outright seizure of their assets
 - C. Personal income taxes levied directly on the regulator's own staff
 - D. Mandatory donations collected annually from international agencies
- Answer: A. One-time revenues from auctions and continuing revenue from annual licence fees**

93. Consumer-protection conditions in telecom licences commonly address matters such as:

- A. Billing practices, complaint mechanisms and mandatory services like emergency calls ✓**
 - B. The detailed internal share structure of the operator's holding company
 - C. The personal credit ratings of each of the company's individual board directors
 - D. The choice of which external auditors are used by the regulator itself
- Answer: A. Billing practices, complaint mechanisms and mandatory services like emergency calls**

94. By clearly defining the rights and obligations of the operator and the regulator, a licence advances the objective of:

- A. Regulatory certainty, which builds confidence and helps attract investment ✓**
 - B. Spectrum refarming carried out across all competing operators
 - C. Eliminating the regulator's discretion entirely and permanently
 - D. Guaranteeing every single operator an exactly equal market share
- Answer: A. Regulatory certainty, which builds confidence and helps attract investment**

95. Radio spectrum is a scarce resource whose international allocation is governed by rules intended to maximise its use while minimising cross-border interference. These rules are managed by the:

- A. International Telecommunication Union (ITU) ✓**
 - B. World Trade Organization (WTO)
 - C. Economic Community of West African States (ECOWAS)
 - D. Individual operators within each country
- Answer: A. International Telecommunication Union (ITU)**

96. Although telephone numbers are in principle infinite, they are treated as a scarce resource partly because:

- A. There are physically only a very small, fixed number of usable digits in existence worldwide

B. Each individual telephone number requires its own permanently dedicated radio frequency to function

C. A given telephone number can only ever be assigned to and used by one single country worldwide

D. People can hold only a limited number of digits in short-term memory, and numbers must be allocated in meaningful blocks ✓

Answer: D. People can hold only a limited number of digits in short-term memory, and numbers must be allocated in meaningful blocks

97. Number portability is a mechanism by which:

A. The operator permanently and exclusively owns every telephone number that it ever issues

B. The subscriber, rather than the operator, owns the number and can take it when switching providers ✓

C. Numbers are periodically auctioned off to whichever operator happens to bid the highest

D. Subscribers must always change their number each time they switch to a completely new provider

Answer: B. The subscriber, rather than the operator, owns the number and can take it when switching providers

98. In allocating spectrum, a regulator may auction it to the highest bidder or allocate it at low cost. Allocating spectrum at low cost is typically intended to:

A. Reduce prices or encourage the rollout of new services ✓

B. Maximise the one-time revenue collected by the government

C. Guarantee that only one operator can ever use the band

D. Remove the need to manage interference between users

Answer: A. Reduce prices or encourage the rollout of new services

99. A primary reason spectrum must be allocated through agreements such as licences, rather than used freely, is to:

A. Increase the total number of digits available for telephone numbers nationally

B. Allow every single licensed operator to broadcast on any frequency it wishes

C. Remove the need for any international coordination of frequencies at all

D. Ensure it is properly distributed and that users do not interfere with one another ✓

Answer: D. Ensure it is properly distributed and that users do not interfere with one another

100. The core principle of interconnection is that:

A. Each operator should deliberately keep its subscribers isolated from rivals' networks

B. Only operators of a roughly similar size are ever legally required to interconnect

C. Interconnection is legally required only between the largest international carriers

D. Everyone should be able to communicate with everyone else, regardless of which network they subscribe to ✓

Answer: D. Everyone should be able to communicate with everyone else, regardless of which network they subscribe to

Part 2

1. When the price of a mobile voice call rises by 10 percent and, as a result, the demand for SMS messages increases by 15 percent, the cross-price elasticity of demand between calls and SMS is positive. This indicates the two services are:

A. Complementary goods — consumers buy more of both when the price of calls rises

B. Substitute goods — consumers switch from calls to SMS as calls become more expensive ✓

C. Independent goods — the demand for SMS is unrelated to the price of calls

D. Inferior goods — the demand for SMS falls as consumer incomes rise

Answer: B. Substitute goods — consumers switch from calls to SMS as calls become more expensive

2. When consumer incomes rise by 10 percent and demand for broadband services rises by 18 percent, the income elasticity of demand for broadband is greater than one. This classifies

broadband as:

- A. A luxury (income-elastic) good, whose demand rises proportionally faster than income ✓**
- B. A necessity (income-inelastic) good, whose demand barely changes with income
- C. An inferior good, whose overall demand actually falls whenever income rises
- D. A Giffen good, whose quantity demanded oddly rises as its own price increases

Answer: A. A luxury (income-elastic) good, whose demand rises proportionally faster than income

3. In a competitive telecom market, consumer surplus represents:

- A. The extra profit that a firm earns by charging a price that is well above its own costs
- B. The difference between what consumers are willing to pay for a service and what they actually pay ✓**
- C. The overall reduction in quantity demanded that is caused by a sudden increase in price
- D. The specific share of the tax burden that is borne by consumers after a new excise tax

Answer: B. The difference between what consumers are willing to pay for a service and what they actually pay

4. When a monopolist restricts output below the socially efficient level and raises the price, the economic loss to society compared with the competitive outcome is called:

- A. Deadweight loss ✓**
- B. Consumer surplus
- C. Producer surplus
- D. Tax incidence

Answer: A. Deadweight loss

5. A mobile platform operator charges low prices to consumers to attract a large user base, and then charges higher prices to businesses that wish to reach those consumers through advertising or app stores. This business model is characteristic of:

- A. A perfectly competitive market in which every single firm charges exactly the same price
- B. A natural monopoly that must be tightly regulated in order to prevent excessive pricing
- C. A two-sided (or multi-sided) market, where the platform intermediates two distinct customer groups ✓**
- D. A command economy where the state directly sets prices for different consumer groups

Answer: C. A two-sided (or multi-sided) market, where the platform intermediates two distinct customer groups

6. A social messaging application becomes more valuable to each new user as more people already use it, because more connections are possible. Economists call this phenomenon:

- A. A positive network externality (or network effect) ✓**
- B. A negative externality, where each new user imposes costs on existing users
- C. Economies of scale, where average costs fall as output rises
- D. Price discrimination, where different users are charged different prices

Answer: A. A positive network externality (or network effect)

7. A mobile subscriber who has accumulated loyalty points, contacts stored in a proprietary format, and a memorable number that cannot be easily ported faces high switching costs. High switching costs benefit the incumbent operator primarily by:

- A. Lowering the average operational cost of serving each and every individual subscriber
- B. Increasing the incentive for entirely new firms to aggressively enter the market
- C. Reducing the competitive pressure it faces, since subscribers find it costly to move to rivals ✓**
- D. Automatically transferring all existing subscribers over to competing rival networks

Answer: C. Reducing the competitive pressure it faces, since subscribers find it costly to move to rivals

8. A fixed-line operator has laid fibre-optic cable across an entire city. This investment is largely non-recoverable if the operator exits the market. Such costs are termed:

- A. Variable costs, which change with the volume of traffic carried

- B. Opportunity costs, which represent the next-best foregone alternative
- C. External costs, which are imposed on parties outside the transaction

D. Sunk costs ✓

Answer: D. Sunk costs

9. A firm whose average cost per subscriber falls steadily as it adds more subscribers is said to exhibit:

- A. Diseconomies of scale, where average cost rises as output increases
- B. Economies of scale ✓**
- C. Constant returns to scale, where average cost does not change with output
- D. Diminishing marginal utility, where each extra unit provides less satisfaction

Answer: B. Economies of scale

10. In a contestable market, even a single firm may price competitively because:

- A. The government directly fixes all market prices at the competitive level by official decree
- B. The incumbent operator voluntarily limits its own profits purely out of social responsibility
- C. Sunk costs throughout the industry are so high that entry and exit are both entirely impossible
- D. The threat of potential entry by rivals constrains the incumbent from raising prices above competitive levels ✓**

Answer: D. The threat of potential entry by rivals constrains the incumbent from raising prices above competitive levels

11. A telecommunications firm that charges each customer exactly their maximum willingness to pay, thereby extracting all consumer surplus, is practising:

- A. Second-degree price discrimination, where prices vary by volume consumed
- B. Third-degree price discrimination, where prices vary by customer group
- C. First-degree (or perfect) price discrimination ✓**
- D. Predatory pricing, where prices are set below cost to drive rivals out

Answer: C. First-degree (or perfect) price discrimination

12. A mobile operator offers the same data plan at one price to residential customers and a lower price to students, based on the observation that students are more price-sensitive. This is an example of:

- A. Third-degree price discrimination, segmenting customers by group elasticity ✓**
- B. First-degree price discrimination, where each buyer pays their unique reservation price
- C. Second-degree price discrimination, where prices depend on quantity purchased
- D. Uniform pricing, because the plan content is identical

Answer: A. Third-degree price discrimination, segmenting customers by group elasticity

13. A broadband provider charges subscribers a fixed monthly access fee and then an additional per-gigabyte usage charge on top. This pricing structure is called:

- A. A flat rate, where a single monthly fee covers all usage
- B. Predatory pricing, aimed at driving rivals from the market
- C. Resale pricing, where the provider buys wholesale and marks up the price
- D. A two-part tariff ✓**

Answer: D. A two-part tariff

14. The Lerner index measures market power as (Price minus Marginal Cost) divided by Price. A Lerner index of zero indicates:

- A. No market power — the firm is pricing at marginal cost, consistent with perfect competition ✓**
- B. Maximum market power — the firm is pricing infinitely far above marginal cost
- C. A natural monopoly — only one efficient firm can ever serve the whole market
- D. Government-mandated pricing — all prices are set directly by regulatory decree

Answer: A. No market power — the firm is pricing at marginal cost, consistent with perfect competition

15. Allocative efficiency in a market is achieved when:

- A. A single dominant firm is large enough to minimise its own average cost of production
- B. All producers throughout the industry use the most technically advanced equipment available
- C. The government directly controls the price of every single product across the whole economy
- D. Goods and services are produced in the quantities that consumers value them, at prices that reflect true costs ✓**

Answer: D. Goods and services are produced in the quantities that consumers value them, at prices that reflect true costs

16. An operator is productively efficient when it:

- A. Sets its retail prices exactly equal to those charged by its nearest competitor
- B. Allocates its output strictly in proportion to the annual incomes of its individual customers
- C. Earns a rate of return that is exactly equal to the prevailing market rate of interest
- D. Produces its output at the lowest possible cost, given current technology and input prices ✓**

Answer: D. Produces its output at the lowest possible cost, given current technology and input prices

17. A regulatory policy that sacrifices short-term competition concerns in order to encourage investment in next-generation infrastructure is prioritising:

- A. Dynamic efficiency — the ability of the market to innovate and improve over time ✓**
- B. Allocative efficiency — matching quantities produced to consumer preferences
- C. Productive efficiency — minimising the cost of current output
- D. Regulatory capture — the regulator acting in the interest of the regulated firm

Answer: A. Dynamic efficiency — the ability of the market to innovate and improve over time

18. A mobile market with three operators, each aware that its pricing decisions will provoke a response from the other two, is best described as an:

- A. Perfect competition, where no single firm influences price
- B. Oligopoly ✓**
- C. Natural monopoly, where a single firm serves the whole market at least cost
- D. Monopsony, where a single buyer faces many sellers

Answer: B. Oligopoly

19. An incumbent operator temporarily prices a service below its cost of provision with the intention of forcing a new entrant out of the market, after which it plans to raise prices. This behaviour is known as:

- A. Penetration pricing, aimed at legitimately building market share
- B. Marginal cost pricing, required of regulated dominant firms
- C. Predatory pricing ✓**
- D. Price discrimination, charging different buyers different prices

Answer: C. Predatory pricing

20. A telecom operator sells broadband, mobile and television services together at a combined price lower than the sum of their individual prices, requiring customers to subscribe to all three. This commercial strategy is called:

- A. Disaggregated resale, where services are sold separately at wholesale
- B. Co-location, where competitors share the operator's physical infrastructure
- C. Resale, where an operator buys wholesale services and retails them
- D. Bundling ✓**

Answer: D. Bundling

21. Regulators use the Herfindahl-Hirschman Index (HHI) to measure market concentration. An HHI approaching 10,000 indicates:

- A. A perfectly competitive market with many equal-sized firms
- B. A market in which the two largest firms each hold exactly 50% of shares
- C. A highly concentrated market approaching a single-firm monopoly ✓**
- D. A market too fragmented to warrant any regulatory oversight

Answer: C. A highly concentrated market approaching a single-firm monopoly

22. In mobile markets, spectrum licences granted to a limited number of operators create a barrier to entry because:

- A. Spectrum licences automatically and permanently grant their holder a legal monopoly
- B. Any firm may use spectrum completely freely once it simply registers with the ITU
- C. Spectrum is generally treated as an unlimited resource that requires no active management
- D. A new firm cannot launch a mobile network without spectrum, which is scarce and tightly regulated ✓**

Answer: D. A new firm cannot launch a mobile network without spectrum, which is scarce and tightly regulated

23. Under rate-of-return regulation the regulator sets the allowed profit level, while under price-cap regulation it sets a ceiling on prices. A main advantage of price-cap regulation over rate-of-return is that it:

- A. Gives the firm an incentive to cut costs, since it keeps any efficiency gains as extra profit ✓**
- B. Guarantees that the regulated firm will always earn at least its own cost of capital
- C. Eliminates the need for the regulator to ever collect any detailed financial data
- D. Prevents any variation in prices at all, thereby providing consumers with full certainty

Answer: A. Gives the firm an incentive to cut costs, since it keeps any efficiency gains as extra profit

24. A mobile operator that also owns the towers its competitors must use is said to be vertically integrated. A regulatory concern with vertical integration is that the firm may:

- A. Be entirely unable to price its own retail services competitively without government subsidies
- B. Favour its own downstream retail operations by providing rivals with degraded or overpriced access to the towers ✓**
- C. Automatically and permanently reduce the overall quality of all of its own retail services
- D. Automatically lose absolutely all of its market power the moment it becomes vertically integrated

Answer: B. Favour its own downstream retail operations by providing rivals with degraded or overpriced access to the towers

25. Economic theory defines a public good as one that is non-rival and non-excludable. A broadcast radio signal most closely resembles:

- A. A public good ✓**
- B. A private good, because the broadcaster can charge for each listener
- C. A club good, because it is non-rival but perfectly excludable
- D. A common resource, because use by one person diminishes supply for others

Answer: A. A public good

26. Structural separation in a telecom market means:

- A. The telecom regulator itself formally divides its own internal functions into several entirely separate government departments
- B. All competing operators active in the market are required to fully share their customer databases with one another at all times
- C. The monopoly infrastructure (such as the local loop) is owned by a separate company from the retail service providers that use it ✓**
- D. The government privatises the incumbent operator by selling off all of its shares directly to members of the general public

Answer: C. The monopoly infrastructure (such as the local loop) is owned by a separate company from the retail service providers that use it

27. Accounting separation requires a vertically integrated operator to:

- A. Physically and legally split its wholesale and retail operations into two entirely separate, independently managed companies
- B. Keep separate financial accounts for its wholesale network and its retail service divisions, as though they were independent entities ✓**
- C. Share the whole of its detailed internal financial accounts freely with every single one of its many

rival operators

D. File completely identical financial statements for absolutely every single country in which it happens to operate

Answer: B. Keep separate financial accounts for its wholesale network and its retail service divisions, as though they were independent entities

28. A universal service obligation (USO) in telecommunications is best defined as:

A. A requirement that every single licensed operator offer every service it currently provides completely free of charge to all

B. An international treaty that permanently and completely prevents any signatory country from restricting its own telecom market

C. A regulatory requirement that certain basic services are made available to all citizens at an affordable price, regardless of location ✓

D. A licence condition that strictly and permanently prohibits any dominant firm from ever raising its own retail prices

Answer: C. A regulatory requirement that certain basic services are made available to all citizens at an affordable price, regardless of location

29. When the provision of universal service is commercially unviable in high-cost rural areas, regulators commonly establish:

A. A regulated price floor that prevents operators from discounting their services in profitable urban areas

B. A mandatory transfer of spectrum rights from profitable operators to unprofitable rural operators

C. A direct legal obligation requiring consumers themselves to subsidise rural service providers

D. A universal service fund financed by levies on operators in proportion to their revenues ✓

Answer: D. A universal service fund financed by levies on operators in proportion to their revenues

30. Market failure due to information asymmetry occurs when:

A. Both buyers and sellers happen to have exactly identical information about the product's quality

B. One side of a market has substantially more information than the other, preventing efficient transactions ✓

C. The government itself has more information than the market does about the optimal quantities to produce

D. A single firm has permanently lower costs than its rivals due to proprietary technology it owns

Answer: B. One side of a market has substantially more information than the other, preventing efficient transactions

31. A positive externality exists when:

A. A transaction between a buyer and seller generates benefits for third parties who are not part of the transaction ✓

B. A firm knowingly imposes ongoing costs on the whole of society that it does not itself ever bear or pay for

C. A government tax deliberately corrects an existing, persistent over-supply of some particular harmful good

D. A monopoly firm simply charges its customers a retail price that sits above the competitive market price

Answer: A. A transaction between a buyer and seller generates benefits for third parties who are not part of the transaction

32. Ex-ante regulation differs from ex-post regulation in that ex-ante:

A. Intervention by the regulator occurs only after a breach has already been detected and formally proven

B. Regulators choose to rely exclusively on general competition law rather than any sector-specific rules

C. Firms remain entirely free to set their own commercial terms until a formal complaint is eventually received

D. Rules are set in advance to shape market behaviour before problems arise, rather than reacting to them after the fact ✓

Answer: D. Rules are set in advance to shape market behaviour before problems arise, rather than reacting to them after the fact

33. An operator designated as having Significant Market Power (SMP) is one that:

- A. Simply holds an operating licence that happens to run for a term of fewer than five years in total duration
- B. Enjoys a position of economic strength that allows it to behave to an appreciable extent independently of competitors, customers and consumers ✓**
- C. Has annual revenues that consistently fall below the national regulatory authority's own official reporting threshold
- D. Operates exclusively within remote rural areas where it faces absolutely no urban competition whatsoever from any rival

Answer: B. Enjoys a position of economic strength that allows it to behave to an appreciable extent independently of competitors, customers and consumers

34. Before assessing whether an operator has SMP, a regulator must first define the relevant market. The relevant product market is defined as:

- A. All of the products offered by every single licensed operator throughout the whole country
- B. Whatever products the regulator personally decides to regulate for its own policy reasons
- C. The smallest set of products that are close enough substitutes to constrain each other's prices ✓**
- D. Only the services that are covered by the dominant operator's own individual licence

Answer: C. The smallest set of products that are close enough substitutes to constrain each other's prices

35. A 'bottleneck' facility in telecommunications is one that:

- A. Is simply located at the exact geographic centre of the national telecommunications network as a whole
- B. Can only ever be used by the incumbent operator under the specific terms written into its own licence
- C. Is shared voluntarily by all of the operators in the market under a formal co-regulation agreement
- D. Is essential for reaching end customers, and which competitors cannot duplicate economically, giving the owner market power ✓**

Answer: D. Is essential for reaching end customers, and which competitors cannot duplicate economically, giving the owner market power

36. Local loop unbundling (LLU) allows:

- A. Alternative operators to rent the physical copper or fibre pair between the exchange and a customer's premises from the incumbent, at regulated prices ✓**
- B. The incumbent operator to freely use rival operators' own transmission equipment entirely free of any charge whatsoever
- C. Individual consumers to personally and permanently own the physical cable that connects their own premises to the exchange
- D. Operators to resell each other's retail services to customers without ever having to own any network infrastructure of their own at all

Answer: A. Alternative operators to rent the physical copper or fibre pair between the exchange and a customer's premises from the incumbent, at regulated prices

37. Tariff rebalancing in the telecom sector typically refers to:

- A. Increasing every single category of retail price uniformly across the board simply to reflect the annual rate of inflation
- B. Transferring spectrum licence charges away entirely from mobile operators and onto fixed-line operators instead, without exception
- C. Adjusting prices so that each service covers its cost, reducing cross-subsidies — for instance, raising local call prices and lowering international prices ✓**
- D. Setting completely identical retail prices for every operator regardless of each one's own underlying cost structure or size

Answer: C. Adjusting prices so that each service covers its cost, reducing cross-subsidies — for

instance, raising local call prices and lowering international prices

38. An operator is likely to retain structural dominance even after market liberalisation largely because it:

- A. Has already lost the great majority of its existing subscribers to newly entering rival firms in the market
- B. Controls legacy infrastructure (such as the local loop or backhaul) that new entrants cannot easily or economically replicate ✓**
- C. Holds an operating licence that just happened to be issued only after the wider market was first opened
- D. Offers absolutely no retail services of any kind at all in the newly opened, fully competitive market

Answer: B. Controls legacy infrastructure (such as the local loop or backhaul) that new entrants cannot easily or economically replicate

39. A regulatory impact assessment (RIA) is designed to:

- A. Audit the detailed financial accounts of a regulated company only after it has already committed a breach
- B. Determine the precise size of the fine to impose on an operator that has been found guilty of market abuse
- C. Estimate the costs and benefits of a proposed regulatory intervention before it is implemented, to ensure its net effect is positive ✓**
- D. Formally replace the entire public consultation process with a single, short written internal report instead

Answer: C. Estimate the costs and benefits of a proposed regulatory intervention before it is implemented, to ensure its net effect is positive

40. A light-touch regulatory approach is typically characterised by:

- A. Minimal intervention, with the regulator stepping in only when there is clear market failure or a specific complaint ✓**
- B. Detailed, highly prescriptive rules that govern every single aspect of an operator's day-to-day behaviour
- C. The outright nationalisation of the dominant operator, carried out purely to prevent any future price abuses
- D. Prohibiting all forms of competition entirely, purely in order to protect the existing natural monopoly

Answer: A. Minimal intervention, with the regulator stepping in only when there is clear market failure or a specific complaint

41. Functional separation requires a dominant operator to:

- A. Sell off its entire network infrastructure outright to a genuinely independent, unrelated third-party company at market value
- B. Provide exactly identical wholesale prices to retail rivals and to its own retail division well before any formal separation occurs
- C. Establish a completely separate universal service fund that is financed entirely from its own revenues alone, with no outside support
- D. Operate its wholesale access network as a functionally separate business unit with independent management and systems, even if ownership remains unified ✓**

Answer: D. Operate its wholesale access network as a functionally separate business unit with independent management and systems, even if ownership remains unified

42. The main argument in favour of infrastructure competition (multiple physical networks) over service competition (multiple retailers on one network) is that it:

- A. Leads to greater dynamic efficiency and innovation as firms compete across multiple layers of the value chain ✓**
- B. Reduces the total overall cost of building broadband networks across the entire country
- C. Is always readily achievable even in rural areas where only a single network is commercially viable
- D. Eliminates absolutely any need for access regulation once the physical networks have been built

Answer: A. Leads to greater dynamic efficiency and innovation as firms compete across multiple layers of the value chain

43. Regulators have intervened in international roaming markets because:

- A. Roaming agreements freely negotiated between operators are explicitly and permanently prohibited under the WTO BTA
- B. Roaming charges were often far above cost, exploiting consumers who had no practical ability to switch provider while abroad ✓**
- C. The ITU formally and strictly requires that all international calls be routed through a single central global hub
- D. Overall consumer demand for roaming services was historically simply too low to justify any commercial provision

Answer: B. Roaming charges were often far above cost, exploiting consumers who had no practical ability to switch provider while abroad

44. When a regulator performs a cost-benefit analysis of a proposed remedy, it should compare:

- A. The regulator's own administrative budget against the total annual revenue of the dominant operator
- B. The average income of rural subscribers against the average income of urban subscribers
- C. The current market share held by each operator against the overall national average
- D. The direct costs imposed on industry and consumers against the economic benefits of correcting the market failure ✓**

Answer: D. The direct costs imposed on industry and consumers against the economic benefits of correcting the market failure

45. A multi-sector regulator that oversees telecoms, energy and water simultaneously may have the advantage of:

- A. Focusing its exclusive expertise narrowly on just a single industry's own unique technical characteristics
- B. Having a considerably larger administrative staff available to investigate each and every individual complaint
- C. Operating independently of government in a way that separate, sector-specific bodies simply never can
- D. Applying consistent regulatory principles across sectors and avoiding conflicts between sector-specific agencies ✓**

Answer: D. Applying consistent regulatory principles across sectors and avoiding conflicts between sector-specific agencies

46. When two large mobile operators propose to merge, reducing the number of national operators from four to three, the regulator's primary competition concern is likely to be:

- A. An immediate and unavoidable increase in spectrum licence costs for the newly merged entity
- B. A risk that the newly merged entity will be entirely unable to recover its licence costs
- C. A reduction in competitive pressure that could lead to higher prices or lower investment ✓**
- D. A formal requirement that the smaller merged operator buy out the regulator itself

Answer: C. A reduction in competitive pressure that could lead to higher prices or lower investment

47. The main advantage of sector-specific telecom regulation over reliance on general competition law is that:

- A. Sector-specific rules can be calibrated to the industry's technical characteristics and applied proactively, without waiting for a market abuse to occur ✓**
- B. Competition law always reaches exactly the same practical outcome as dedicated sector-specific regulation would in every case
- C. Sector regulation is inherently more flexible simply because it has no prescribed rules of any kind at all to follow
- D. Competition authorities are formally and permanently forbidden from ever intervening in telecom markets under any circumstances

Answer: A. Sector-specific rules can be calibrated to the industry's technical characteristics and applied proactively, without waiting for a market abuse to occur

48. Asymmetric regulation imposes obligations on the dominant operator that are not applied to smaller rivals. The justification for this asymmetry is that:

A. Smaller operators are simply managed better overall and therefore need considerably fewer rules imposed on them

B. Only the dominant operator has the market power to harm competition or consumers in ways that the market itself cannot correct ✓

C. All operators active in the market should be regulated in exactly the same way regardless of their market share

D. Asymmetric regulation of this specific kind is required by the WTO Reference Paper in absolutely all cases

Answer: B. Only the dominant operator has the market power to harm competition or consumers in ways that the market itself cannot correct

49. Periodic market reviews, in which the regulator reassesses which operators have SMP and what remedies apply, serve primarily to:

A. Permanently and steadily increase the regulatory obligations placed on all operators in the market over time

B. Allow the regulator to raise additional annual licence fees specifically from the dominant operators alone

C. Ensure that regulatory obligations remain proportionate as market conditions evolve, and are removed when no longer needed ✓

D. Transfer ownership of the incumbent operator back to the state if effective competition ever fails to develop

Answer: C. Ensure that regulatory obligations remain proportionate as market conditions evolve, and are removed when no longer needed

50. Regulatory forbearance occurs when a regulator:

A. Simply fails to apply its own regulations properly due to administrative error or chronic under-resourcing

B. Is formally prevented from intervening at all by a court injunction that was obtained by the operator itself

C. Deliberately increases the overall stringency of its regulation in response to changing market conditions

D. Deliberately refrains from exercising its available powers in a market, on the grounds that competition is sufficient ✓

Answer: D. Deliberately refrains from exercising its available powers in a market, on the grounds that competition is sufficient

51. Regulatory independence from the regulated industry encompasses institutional, financial and operational dimensions. Which dimension is concerned with ensuring regulators are not housed within the same organisation as operators?

A. Financial independence

B. Institutional independence ✓

C. Operational independence

D. Political independence

Answer: B. Institutional independence

52. A key risk where the regulator's annual budget is set entirely at the government's discretion is that:

A. The regulator will often over-spend heavily on enforcement against politically connected firms

B. Operators will demand the right to audit the regulator's own accounts directly and regularly

C. The regulator may be starved of resources as an indirect way of constraining its activity ✓

D. The regulator will automatically become independent of both the government and the operators

Answer: C. The regulator may be starved of resources as an indirect way of constraining its activity

53. Which of the following is a mechanism that makes a regulator accountable without compromising its independence?

A. A requirement to publish an annual report explaining its decisions to the legislature and public ✓

B. Allowing the responsible minister to personally reverse any decision that the regulator makes

- C. Requiring the regulated operator itself to formally sign off on all regulatory decisions made
- D. Permitting the regulator to operate freely without any published criteria of any kind at all

Answer: A. A requirement to publish an annual report explaining its decisions to the legislature and public

54. Transparency in regulation requires that:

- A. All correspondence between the regulator and the operators it regulates must be kept strictly confidential at all times
- B. The regulator shares only its bare final decisions with the public, without ever publishing any of its underlying reasoning
- C. The criteria, process and reasoning behind regulatory decisions are publicly available so that affected parties can understand and challenge them ✓**
- D. Consultation documents produced by the regulator are circulated only ever to the single dominant operator in the market

Answer: C. The criteria, process and reasoning behind regulatory decisions are publicly available so that affected parties can understand and challenge them

55. Regulatory certainty is valued by investors primarily because:

- A. Certain, stable rules always and automatically lead to permanently lower retail prices for all consumers
- B. Predictable rules reduce the risk premium they require when committing long-term capital to network infrastructure ✓**
- C. Regulatory certainty entirely prevents any operator from ever being able to enter or exit the market
- D. Certain rules completely eliminate the need for any formal licencing process to exist at all

Answer: B. Predictable rules reduce the risk premium they require when committing long-term capital to network infrastructure

56. Under RPI-X price-cap regulation, if RPI is the retail price index and X is an efficiency factor, operators are required to:

- A. Increase their regulated prices by the full RPI each year regardless of their own internal cost savings
- B. Keep all of their regulated prices fixed in nominal terms indefinitely, regardless of ongoing inflation
- C. Reduce their regulated prices in real terms by X per cent per year, passing efficiency gains to consumers ✓**
- D. Set their prices at exactly the rate of inflation minus whatever government subsidy happens to apply

Answer: C. Reduce their regulated prices in real terms by X per cent per year, passing efficiency gains to consumers

57. A high X factor in a price-cap formula (RPI-X) implies:

- A. The regulator expects the operator to achieve large efficiency savings and requires it to pass them on to consumers quickly ✓**
- B. The regulator simply expects the wider market to become steadily less competitive over the coming years
- C. The regulated operator will instead receive a direct subsidy equal to X per cent of its total annual revenues
- D. The entire price cap will automatically be removed once exactly X years of regulation have fully passed

Answer: A. The regulator expects the operator to achieve large efficiency savings and requires it to pass them on to consumers quickly

58. A well-known drawback of rate-of-return (RoR) regulation is that it may create an incentive for the regulated firm to:

- A. Under-invest entirely in new infrastructure simply in order to avoid attracting any regulatory scrutiny
- B. Reduce its retail prices below cost specifically in order to deliberately drive its rivals out of the market
- C. Resist any and all technological change that would require it to undertake new capital spending
- D. Over-invest in capital (the Averch-Johnson effect), because profit is calculated as a percentage of the asset base ✓**

Answer: D. Over-invest in capital (the Averch-Johnson effect), because profit is calculated as a percentage of the asset base

59. A licence condition requiring a mobile operator to achieve a minimum 95 percent population coverage and a maximum 2-second call-setup delay is an example of:

- A. Quality-of-service (QoS) regulation imposing minimum performance standards ✓**
- B. Price-cap regulation limiting how much the operator may charge
- C. Accounting separation requiring separate financial records
- D. Structural separation requiring distinct ownership of the access network

Answer: A. Quality-of-service (QoS) regulation imposing minimum performance standards

60. A regulator that requires a dominant operator to submit annual regulatory accounts broken down by network element cost, retail revenue and margin is using:

- A. Accounting separation to improve cost transparency and detect potential cross-subsidies ✓**
- B. Structural separation to legally require the creation of two entirely distinct companies
- C. A formal price-cap review carried out specifically to determine the operator's new X factor
- D. A formal market review carried out specifically to define the relevant product market

Answer: A. Accounting separation to improve cost transparency and detect potential cross-subsidies

61. A regulatory sandbox allows:

- A. Operators to voluntarily exclude themselves from all forms of regulation for a fixed set period of time
- B. Governments to informally test new regulatory agencies before they are ever formally established in law
- C. Firms to test innovative products or business models in a controlled environment with relaxed rules, subject to safeguards ✓**
- D. Regulators to freely impose considerably higher fines than the underlying law would normally ever permit

Answer: C. Firms to test innovative products or business models in a controlled environment with relaxed rules, subject to safeguards

62. A sunset clause in a regulatory instrument provides that:

- A. The regulated firm concerned must fully upgrade its network before a specified statutory deadline arrives
- B. The sitting regulator personally must resign from office once its own five-year term formally ends
- C. All obligations under the licence are instead phased in gradually over a period of several years
- D. The regulation automatically expires after a set time unless the regulator actively decides to renew it ✓**

Answer: D. The regulation automatically expires after a set time unless the regulator actively decides to renew it

63. During a periodic market review, if the regulator finds that effective competition has developed and the dominant operator no longer has SMP, the appropriate response is to:

- A. Remove or relax the sector-specific obligations previously imposed on that operator ✓**
- B. Impose considerably stricter conditions specifically to safeguard the competition that has newly emerged
- C. Transfer the operator's own spectrum holdings directly to a separate, state-owned entity instead
- D. Launch a full formal inquiry to determine whether outright re-nationalisation of the firm is needed

Answer: A. Remove or relax the sector-specific obligations previously imposed on that operator

64. When an interconnection agreement cannot be reached through commercial negotiation, most regulatory frameworks allow either party to:

- A. Withdraw services to the rival operator until an agreement is reached
- B. Refer the dispute to the regulator for binding arbitration or determination ✓**
- C. Seek damages from the rival's shareholders in a consumer court
- D. Delay interconnection indefinitely without incurring any penalty

Answer: B. Refer the dispute to the regulator for binding arbitration or determination

65. Regulatory risk, from the perspective of an investor in telecoms infrastructure, refers to:

- A. The physical risk that network equipment might be seriously damaged by extreme weather events
- B. The risk that the regulator will change the rules in ways that reduce the expected return on invested capital ✓**
- C. The risk that existing subscribers will switch to rival operators once new services are launched
- D. The risk that competition law will legally prohibit the operator from charging cost-based prices

Answer: B. The risk that the regulator will change the rules in ways that reduce the expected return on invested capital

66. Monitoring and enforcement are distinct regulatory activities. Monitoring is primarily concerned with:

- A. Imposing fines and other penalties on firms that have already formally breached their own licences
- B. Deciding in the first place what specific obligations should be imposed on operators generally
- C. Publishing detailed annual reports covering the regulator's own internal financial performance
- D. Collecting data and tracking compliance with regulatory obligations before a breach has occurred ✓**

Answer: D. Collecting data and tracking compliance with regulatory obligations before a breach has occurred

67. A regulator that responds to a minor licence breach with an informal warning, escalating to a formal compliance notice and then a fine if the breach persists, is applying:

- A. A graduated (or proportionate) enforcement approach ✓**
- B. Regulatory capture, whereby the regulator avoids penalising the industry
- C. Co-regulation, where the industry jointly decides on the appropriate penalty
- D. Zero regulation, since no immediate fine is imposed on the first breach

Answer: A. A graduated (or proportionate) enforcement approach

68. A voluntary code of practice adopted by operators without legislative force serves primarily to:

- A. Replace every single one of the formal licence conditions with purely non-binding voluntary commitments instead
- B. Establish baseline standards of consumer treatment that operators commit to publicly, creating reputational incentives for compliance ✓**
- C. Allow the regulator to freely impose fines for any breach whatsoever without ever following any form of due process
- D. Transfer all regulatory responsibility for the matter entirely over to the ordinary courts of law instead

Answer: B. Establish baseline standards of consumer treatment that operators commit to publicly, creating reputational incentives for compliance

69. When a regulator proposes a new interconnection pricing methodology, best practice requires it to publish a consultation document and allow a period for:

- A. The responsible government minister to formally approve the document before it is sent to any operator
- B. The single dominant operator alone to veto any proposals that it personally finds commercially disadvantageous
- C. Operators, consumers and other interested parties to comment before the methodology is finalised ✓**
- D. The courts to formally pre-approve the methodology as being legally valid before it is even published

Answer: C. Operators, consumers and other interested parties to comment before the methodology is finalised

70. In developing economies, one frequently cited obstacle to effective regulation is:

- A. An overall excess of regulatory staff who collectively impose unnecessary administrative burdens on every single licensed operator
- B. The knowledge and information gap between the regulator and the companies it regulates, due to limited technical and financial expertise in the regulatory body ✓**
- C. The outright and total refusal of incumbent operators to obey the law at all, under absolutely any

circumstances whatsoever

D. The complete and total absence of any legislation whatsoever enabling the creation of a regulatory body in the very first place

Answer: B. The knowledge and information gap between the regulator and the companies it regulates, due to limited technical and financial expertise in the regulatory body

71. Why are licence conditions a particularly powerful regulatory instrument compared with general guidelines?

A. Because informal guidelines are only ever published after a court order, while formal licences are much quicker

B. Because binding licence conditions apply automatically to every single firm across every sector of the economy

C. Because non-binding guidelines must first be approved by the ITU, whereas formal licences never need to be

D. Because they are legally binding on the licence holder and breach can result in enforcement action or licence revocation ✓

Answer: D. Because they are legally binding on the licence holder and breach can result in enforcement action or licence revocation

72. A regulatory requirement for operators to maintain a formal written complaint procedure and respond within a set number of days is designed to:

A. Prevent ordinary consumers from ever taking their complaints directly to a court or independent tribunal

B. Ensure consumers have an accessible redress mechanism and operators take service quality seriously ✓

C. Transfer consumer protection responsibilities entirely away from the regulator and over to the operator instead

D. Allow the operator to legitimately charge consumers a fee for the internal processing of their complaints

Answer: B. Ensure consumers have an accessible redress mechanism and operators take service quality seriously

73. Regulators in smaller markets often benchmark their rules and tariff levels against those in comparable countries. The primary purpose of such benchmarking is to:

A. Simply copy the exact rules used by the single largest economy, entirely regardless of local market conditions back at home

B. Entirely replace the whole of the national public consultation process with a purely international one instead altogether

C. Draw on evidence from multiple markets to set well-informed, defensible regulatory parameters in the absence of sufficient domestic data ✓

D. Allow individual operators themselves to freely choose whichever country's regulatory framework they personally would prefer

Answer: C. Draw on evidence from multiple markets to set well-informed, defensible regulatory parameters in the absence of sufficient domestic data

74. In most countries, the telecommunications regulatory framework assigns the legislature (parliament) the role of:

A. Setting all wholesale interconnection prices individually on a strict case-by-case basis each time

B. Enacting the primary legislation that establishes the regulatory body and confers its powers ✓

C. Issuing individual operator licences directly and personally managing all spectrum allocation decisions

D. Investigating each and every individual consumer complaint about operators' billing practices

Answer: B. Enacting the primary legislation that establishes the regulatory body and confers its powers

75. An independent regulatory authority is often governed by a multi-member board rather than a single regulator. The main benefit of a board structure is that:

- A. A multi-member board can issue licences considerably faster because each member simply signs separately
- B. Individual board members are personally and legally liable for regulatory decisions, which deters bias
- C. A multi-member board structure does not require any underlying legislation at all in order to be established

D. Decisions represent a range of views and are less vulnerable to the bias or capture of any single individual ✓

Answer: D. Decisions represent a range of views and are less vulnerable to the bias or capture of any single individual

76. Spectrum refarming refers to the process of:

A. Reassigning spectrum bands previously used by one technology (e.g. 2G) to a newer, more spectrally efficient technology (e.g. 4G or 5G) ✓

- B. Allocating entirely new spectrum bands that were only very recently discovered through satellite observation work
- C. Transferring spectrum rights permanently between two or more entire countries under a formal, signed ITU agreement
- D. Increasing the total amount of MHz available within a given band simply by reducing the size of its guard bands further

Answer: A. Reassigning spectrum bands previously used by one technology (e.g. 2G) to a newer, more spectrally efficient technology (e.g. 4G or 5G)

77. In spectrum secondary trading, spectrum rights are:

- A. Returned automatically to the government for free reallocation once they are no longer needed by anyone
- B. Auctioned directly by the ITU itself to the single highest bidder across every one of its member states

C. Transferred or leased from one licence holder to another through market transactions, subject to regulator approval ✓

- D. Allocated automatically to whichever operator happens to have the longest network operating history

Answer: C. Transferred or leased from one licence holder to another through market transactions, subject to regulator approval

78. Cognitive radio technology enables more efficient spectrum use primarily by:

- A. Physically expanding the total radio spectrum available worldwide by opening up entirely new, previously unused frequency ranges

B. Allowing devices to sense and opportunistically use spectrum bands that are temporarily idle, without causing harmful interference to primary users ✓

- C. Requiring every single operator in the market to permanently share just one licensed frequency band between all of them
- D. Replacing all existing licensed spectrum bands entirely with unlicensed, licence-exempt bands for every kind of telecom service

Answer: B. Allowing devices to sense and opportunistically use spectrum bands that are temporarily idle, without causing harmful interference to primary users

79. A regulator may impose spectrum caps (or spectrum limits) on individual operators in order to:

- A. Guarantee that every single operator in the market ends up holding exactly the same amount of spectrum
- B. Reduce the total overall amount of spectrum that is consumed by all operators across the whole country
- C. Allow the incumbent operator to hold onto all of the remaining spectrum once liberalisation has occurred

D. Prevent any single operator from acquiring so much spectrum that it can disadvantage rivals, particularly in auctions ✓

Answer: D. Prevent any single operator from acquiring so much spectrum that it can disadvantage

rivals, particularly in auctions

80. When a country's radio transmissions could cause interference to services in a neighbouring country, the coordination process is managed through:

- A. Bilateral and multilateral agreements, guided by the ITU Radio Regulations ✓**
- B. The World Trade Organization's Basic Telecommunications Agreement, negotiated between member states
- C. A dedicated telecommunications sub-committee that operates under the UN Security Council itself
- D. The National Regulatory Authority of the affected country acting entirely on its own, unilaterally

Answer: A. Bilateral and multilateral agreements, guided by the ITU Radio Regulations

81. In the E.164 international numbering plan, a full international telephone number comprises:

- A. A network code followed directly by a subscriber number, with an overall minimum of ten digits
- B. A two-letter country prefix followed by both an area code and a separate subscriber number
- C. A country code followed by the national (significant) number, with the total not exceeding 15 digits ✓**
- D. A six-digit operator code followed directly by the individual subscriber's own national number

Answer: C. A country code followed by the national (significant) number, with the total not exceeding 15 digits

82. Non-geographic numbers differ from geographic numbers in that they:

- A. Are not tied to a specific physical location, and are used for services such as freephone, premium-rate, or shared-cost calls ✓**
- B. Can only ever be assigned to mobile subscribers specifically, and never at all to any fixed-line operator
- C. Must always begin with the full international country code of whichever network originally originates the call
- D. Are reserved exclusively and permanently for the sole use of emergency services alone, in every single country

Answer: A. Are not tied to a specific physical location, and are used for services such as freephone, premium-rate, or shared-cost calls

83. A freephone (toll-free) service number is one where:

- A. Neither the caller nor the called party pays anything at all, as the government fully subsidises the call
- B. The caller instead pays a considerably higher-than-standard rate, which the operator itself then retains
- C. The number in question is available only to emergency services and hospitals, and to no one else
- D. The cost of the call is borne by the called party (the business) rather than the calling consumer ✓**

Answer: D. The cost of the call is borne by the called party (the business) rather than the calling consumer

84. Premium-rate service numbers expose consumers to the risk of high unintended charges. A common regulatory response is to:

- A. Prohibit the use of premium-rate numbers entirely and completely across the whole market
- B. Require clear price disclosure before the consumer connects and impose a maximum per-minute or per-call cap ✓**
- C. Require that all premium-rate calls be paid for fully in advance by the calling consumer
- D. Allow operators to freely set their own premium rates without any disclosure obligation whatsoever

Answer: B. Require clear price disclosure before the consumer connects and impose a maximum per-minute or per-call cap

85. Short codes (e.g. 3- to 6-digit numbers) are administered by national regulators because:

- A. Short codes are instead allocated by the ITU to each individual country as part of the wider E.164 plan
- B. They are a scarce resource — their brevity limits how many exist — and must be allocated**

fairly across competing service providers ✓

C. Short codes are effectively infinite in overall supply and therefore require no formal administration at all

D. Short codes can only ever be used by government agencies and by telecom regulators themselves

Answer: B. They are a scarce resource — their brevity limits how many exist — and must be allocated fairly across competing service providers

86. In a standard telephone call, the originating operator is the one that:

A. Receives the call on behalf of the called subscriber and is the one that rings their phone

B. Issues the specific number at which the called party can ultimately be reached by others

C. Carries the call from the calling subscriber and delivers it to the network of the terminating operator ✓

D. Determines the final retail price that is actually charged to the called party themselves

Answer: C. Carries the call from the calling subscriber and delivers it to the network of the terminating operator

87. Mobile termination rates (MTRs) are regulated in many countries because:

A. Mobile operators in general charge their own subscribers far too little money for all of their incoming calls received

B. The ITU formally and explicitly mandates that all mobile termination rates be set at zero in every single member state

C. Mobile termination rates of this kind are only ever paid directly by the government, and are never paid by any operator

D. The terminating operator has a monopoly over access to its own subscribers, allowing it to charge above-cost rates to originating operators ✓

Answer: D. The terminating operator has a monopoly over access to its own subscribers, allowing it to charge above-cost rates to originating operators

88. Long-run incremental cost (LRIC) is used as a basis for regulated interconnection prices because:

A. It simply measures the actual historic accounting cost that the operator has already spent at some point in the past

B. It guarantees the regulated operator a fixed, government-mandated profit margin specifically on interconnection traffic

C. It is simply the specific costing model that the WTO officially prefers for use in all cross-border interconnection

D. It reflects the efficient forward-looking cost of providing the specific service, excluding costs unrelated to the traffic being terminated ✓

Answer: D. It reflects the efficient forward-looking cost of providing the specific service, excluding costs unrelated to the traffic being terminated

89. Retail-minus pricing sets the wholesale access price at:

A. A wholesale access price set quite deliberately well above the incumbent's own retail price in order to deter competitive entry

B. The incumbent's retail price minus the retail costs (marketing, billing, etc.) it avoids when selling wholesale, ensuring viable retail competition ✓

C. Zero, on the basis that all wholesale network access should always be provided completely free of any charge whatsoever to anyone

D. The full and complete cost of building an entirely new, fully competing rival network completely from scratch every single time

Answer: B. The incumbent's retail price minus the retail costs (marketing, billing, etc.) it avoids when selling wholesale, ensuring viable retail competition

90. Domestic roaming agreements allow a mobile operator to:

A. Offer its own subscribers entirely free calls whenever they make any international visit to a partner country

B. Permanently use a foreign operator's own spectrum inside its national borders without any time limit

C. Charge its subscribers a noticeably lower price whenever they happen to be in their home region

D. Provide services to its subscribers in geographic areas it does not cover, by using the network of another domestic operator ✓

Answer: D. Provide services to its subscribers in geographic areas it does not cover, by using the network of another domestic operator

91. Regulators impose wholesale roaming caps on visited-network operators. The effect of a wholesale roaming cap is to:

A. Prevent home-network operators from ever being able to sign any roaming agreements with other operators at all, under any circumstances

B. Allow subscribers to freely use absolutely any network they personally wish, without ever needing any operator permission whatsoever

C. Limit the price that visited-network operators may charge the home-network operator, which can then pass lower retail prices to travelling subscribers ✓

D. Set the exact retail price that is charged to consumers by the home operator directly and entirely unilaterally, without exception

Answer: C. Limit the price that visited-network operators may charge the home-network operator, which can then pass lower retail prices to travelling subscribers

92. A Mobile Virtual Network Operator (MVNO) differs from a full Mobile Network Operator (MNO) in that an MVNO:

A. Owns its own radio spectrum and base stations outright, but simply does not maintain any retail customer base at all

B. Operates exclusively within remote rural areas where full MNOs happen to have absolutely no network coverage whatsoever

C. Is licensed under a general class licence that is not subject to any conditions at all, quite unlike a full MNO

D. Does not own radio spectrum or a radio access network, and instead buys wholesale network access from an MNO to sell services to end customers ✓

Answer: D. Does not own radio spectrum or a radio access network, and instead buys wholesale network access from an MNO to sell services to end customers

93. Passive infrastructure sharing involves operators sharing:

A. Physical structures such as towers, masts, ducts and shelters, without sharing the active radio or transmission equipment ✓

B. The exact same radio frequencies and active antennas installed together on one single, shared cell site

C. Spectrum licences themselves, so that both operators end up transmitting together on one single common frequency band

D. Customer billing systems and subscriber management platforms used internally and separately by each individual operator

Answer: A. Physical structures such as towers, masts, ducts and shelters, without sharing the active radio or transmission equipment

94. Radio Access Network (RAN) sharing goes beyond passive sharing in that operators share:

A. The active radio equipment (antennas, transceivers, controllers) in addition to towers and other physical structures ✓

B. Only the masts and shelters themselves, and not any of the associated electronic equipment at all

C. Core network elements such as the central switching and routing infrastructure used by each operator

D. Spectrum licences, so that every operator taking part in the shared RAN uses one single frequency band

Answer: A. The active radio equipment (antennas, transceivers, controllers) in addition to towers and other physical structures

95. Net neutrality is the principle that:

A. Only officially neutral third parties are ever permitted to operate any internet exchange points at all

B. Internet service providers must offer completely free access to official government websites

specifically and exclusively

C. Internet service providers must treat all internet traffic equally, without discriminating based on source, destination or content type ✓

D. Content companies must instead pay individual consumers directly for the privilege of consuming their own online services

Answer: C. Internet service providers must treat all internet traffic equally, without discriminating based on source, destination or content type

96. In mobile number portability (MNP), the 'losing' operator is:

A. The operator that is actively gaining the subscriber who wishes to keep hold of their existing number

B. The operator from which the subscriber is porting away — the one that currently holds the number ✓

C. The operator responsible for routing the ported call once the entire porting process is complete

D. The operator that centrally administers the national number portability database on behalf of all

Answer: B. The operator from which the subscriber is porting away — the one that currently holds the number

97. Regulators that set maximum porting timelines (e.g. one working day) do so primarily to:

A. Prevent operators from using slow porting processes to discourage subscribers from switching ✓

B. Allow operators to complete all technical work before the customer is aware of the port

C. Ensure that porting requests are always processed in the order received

D. Limit the number of porting requests an operator must handle per month

Answer: A. Prevent operators from using slow porting processes to discourage subscribers from switching

98. A short licence duration (e.g. five years) compared with a long one (e.g. twenty years) gives the regulator more:

A. Greater certainty for investors who specifically need predictable returns over a much longer period

B. More time before any formal regulatory review of the operator's own conduct is ever required

C. Opportunities to revise the licence conditions as market and technology conditions change ✓

D. Additional spectrum allocation rights that can then be distributed out to brand-new entrants

Answer: C. Opportunities to revise the licence conditions as market and technology conditions change

99. When renewing a licence, a regulator may attach new or revised conditions. The main constraint on this power is that:

A. No new conditions of any kind may ever be added to an existing licence once it has originally been granted to any operator at all

B. Only the ITU itself may ever formally approve any new conditions that happen to be added at the time of a licence renewal

C. New conditions must be proportionate, justified by current market circumstances, and the operator must have fair notice and an opportunity to respond ✓

D. The national government treasury must always separately approve each and every single new condition for its own fiscal impact

Answer: C. New conditions must be proportionate, justified by current market circumstances, and the operator must have fair notice and an opportunity to respond

100. A regulatory requirement that all licensed operators connect emergency calls free of charge, even when a subscriber has no credit, is justified on the basis that:

A. Emergency calls actually generate significant revenue that fully offsets their own zero-price status

B. Emergency call access is a fundamental public interest obligation that overrides commercial pricing considerations ✓

C. The WTO's Basic Telecommunications Agreement explicitly prohibits any charge for emergency service calls

D. Subscribers who personally call emergency services are later refunded directly by the government itself

Answer: B. Emergency call access is a fundamental public interest obligation that overrides commercial pricing considerations